AUO CORPORATION

Specification for Approval

INCOMING INSPECTION STANDARD FOR
24" TFT-LCD MODULES

Model: M240UAN02 (200 Series)-Non-glare

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Approved by	Prepared by
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Customer	**Checked and Approved by**

1. **Scope:**

1.1 The incoming inspection standards shall be applied to TFT-LCD Modules (hereinafter called "Modules") that supplied by AU Optronics Corporation (hereinafter called "seller").

1.2 Specifications contains

- Electrical inspection specification
- Appearance specification
- Outside dimension specification

2. **Incoming inspection:**

The buyer (customer) shall inspect the modules within twenty calendar days of the delevery date (the "inspection period") at its own cost. The results of the inspection (acceptance or rejection) shall be recorded in writing, and a copy of this writing will be promptly sent to the seller.

The buyer may, under commercially reasonable reject procedures, reject an entire lot in the delivery involved if, within the inspection period ,such samples of modules within such lot show an unacceptable number of defects in accordance with this incoming inspection standards, provided however that the buyer must notify the seller in writing of any such rejection promptly, and not later than within three business days of the end of the inspection period.

Should the buyer fail to notify the seller within the inspection period, the buyer's right to reject the modules shall be lapsed and the modules shall be deemed to have been accepted by the buyer.

3. Inspection sampling method:

Unless otherwise agree in writing, the method of incoming inspection shall be based on MIL-STD-105E.

3.1 Lot size: Quantity per shipment lot per model.

3.2 Sampling type: Normal inspection, single sampling.

3.3 Sampling level: Level II.

3.4 Acceptable quality level (AQL):

Major defect: AQL= 1.0%.

Minor defect: AQL= 2.5%.

4. Inspection instruments:

4.1 Pattern generator: LD-2000 or equivalent model.

4.2 Video board: AU video board or equivalent. The output of the signal should comply with the specification provided by AU.

4.3 Luminance colorimeter: Topcon BM-7 or equivalent model

5. Inspection environment conditions:

5.1 Room temperature: 20 ~ 25 C

5.2 Humidity: 65 ± 5% RH.

5.3 Illumination: Fluorescent light (day-Light Type) display surface illumination to be 300 ~ 700 Lux. (standard 500Lux.)

5.4 To be a distance about 35±5 cm in front of LCD unit, viewing line should be perpendicular to the surface of the module judge the visual appearance with human's eyes (±30° viewing edge will be allowed).

5.5 Take off the protection film of polarizer while judging the display area.

5.6 If there is any question while judging, check the panel again in operating mode.

6. Classification of defects:

Defects are classified as major defects and minor defects according to the degree of defectiveness defined herein.

Major defects: A major defect is a defect that is likely to result in failure, or to reduce materially the usability of the product for its intended purpose.

Minor defects: A minor defect either is a defect that is not likely to reduce materially the usability of the product for its intended purpose, or is a departure from an established having little bearing on the effective use or operation of the product.

6.1 Electrical inspection specification

	Inspection Item	Specification
1	Line defect	Can’t be seen.
2	Bright dots	≤ 2 dots (note1,4)
3	Dark dots	≤ 5 dots
4	Total dots defect	≤ 5 dots
5	Adjacent dot defect (note 1,2,3)	Two continuous bright dots (vertical, horizontal, oblique): ≤ 1 pair
		Three or more continuous bright dots (vertical, horizontal, oblique): Not allowed
		Two continuous dark dots (vertical, horizontal, oblique): ≤ 2 pair.
		Three or more continuous dots – to be of any combination of dark dot and bright dot (vertical, horizontal, oblique): Not allowed
		Distance between 2 Bright dots: ≥ 15 mm Distance between 2 Dark dots: ≥ 15 mm Distance between Bright and Dark dots: ≥ 15 mm
6	Display non-uniformity Or Mura (Note 5,6)	Use 3% ND filter at L128 Pattern or judged by equivalent limit sample

Note 1: For bright dot defect, bright area should be larger than 1/2 area of a sub-pixel to be count as 1 dot defect. A dot defect that is smaller than the defined dot defect will be treated as small bright dot.

The drawing of 1/2 area sub-pixel definition: The 1/2 area sub-pixel can be defined as below one or more of specific shapes (Fig.1).

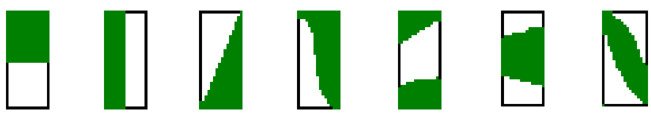


Fig.1

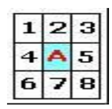


Fig.2

All bright dot defects should not be noticeable by observer under specified inspection environment (Please refer to item 5). Bright dot shall be visible through 8% ND filter.

Note 2: (Fig.2) Adjacent-dot defect (refer to picture, dot 1,2,..,8 around A are all A’s adjacent dots) should be inspected under the same display pattern in any one of White /Black /Red /Green /Blue /Monotone Gray pattern.

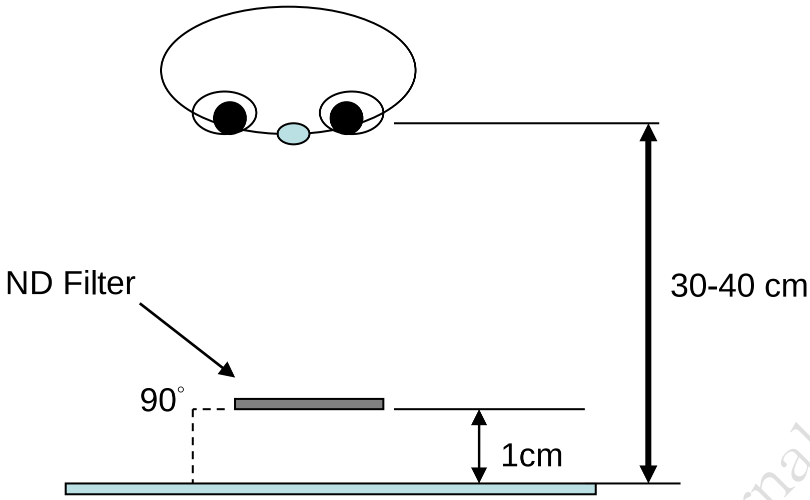
Note 3: Adjacent-dot defect should be observed under any one of White /Black /Red /Green /Blue pattern. 1 pair of bright dots equals 2 dots. Inspection patterns: Standard inspection patterns of dot defect are listed below. AUO uses these patterns as standard criteria for judging dot defect. Please inform AUO if any other pattern is to be used to examine dot defect.

Test Pattern	Defect
Full Black	For bright dot(s)
Full White	For dark dot(s)
Monotone Red /Green /Blue	For bright and dark dot(s)

Note 4: The judgment criteria of particle occurred bright dot is the same as bright dot judgment criteria.

Note 5: The display uniformity (general mura) symptoms will use 3% ND Filter. That is, the definition of Not Noticeable means the symptom can be covered by 3% ND Filter.

Note 6: The inspection method of ND Filter - holding ND filter in front of the panel around 1 cm and examine the panel from 35±5 cm in the front view for 3 seconds.



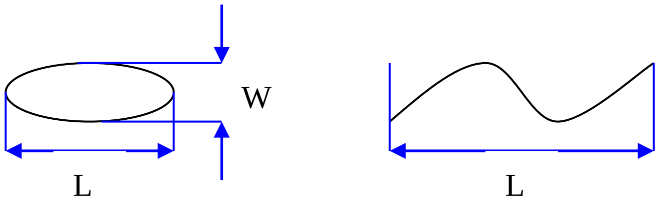
6.2 Appearance inspection specification

Judge area	Judge item		Inspection specification			Judge criterion: Major	Judge criterion: Minor
Active Area (Note 4)	Particles, scratch and bubbles in display area (include bright & dark spot, Note 1, 2, 3)	Round	Average diameter (D) :(mm)		Numbers (N)		❍
			D < 0.25		Disregarded		
			0.25 ≤ D ≤ 0.7		N ≤ 5		
			0.7 < D		N = 0		
		Linear	Width: W (mm)	Length: L (mm)	Numbers		O
			W < 0.1 or L < 2		Disregarded		
			0.1 ≤ W ≤ 0.3 or 2 ≤ L ≤ 10		N ≤ 5		
			W > 0.3 or L >10		N = 0		
BM area polarizer dent (Note 5)	Dent	Round	D ≤ 2.0		N ≤ 2		O
Bezel	Scratch		No harm				O
	Dirt						O
	Wrap		No dangerous			O	
	Sunken		No harm			O	
Label (S/N, B/L, Week code)	No label		No			O	
	Invert label					O	
	Broken					O	
	Dirt		Word can be read.				O
	Not clear						O
	Word out of shape						O
	Mistake		No				O
	Position		Be attached on right position				O

Solder	Appearance	Can’t see the abnormal color, shape, hurt, dirt (fused goods, etc.). If it is necessary, please prepare sample.	O	
Screw	Not enough	No	O	
	Limp	No	O	
Connector	Connection status	Need correct connection.	O	
FPC/FFC	Broken	No	O	

Note 1: When L>=2W, defect count as liner defect.

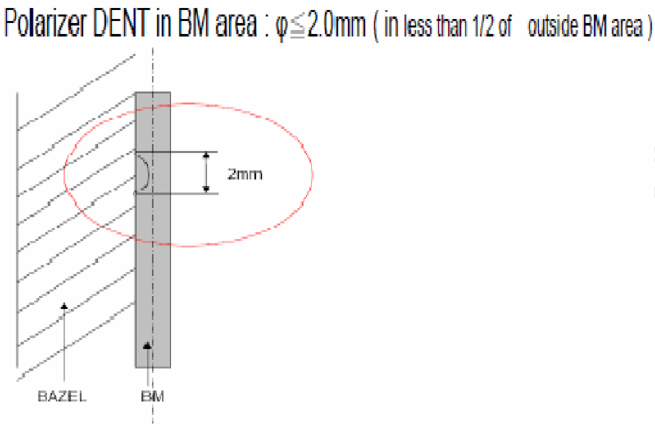
Note 2: D=1/2(W+L)



Note 3: To verify the responsbility of following defects was caused by supplier, the IQC checks as requested on above items before mass production such as the Polarizer Scratch, Gap Mura, TFT Glass broken…etc.

Note 4: This item describe the defect found in the active (display) area.
The descrpition applies to defect that actually take place in TFT, polarizer, or B/L.

Note 5: BM area polarizer dent. ($\varphi \leqq 2.0$mm (inside1/2 width of BM area))



7. Inspection judgement:

7-1 The judgement of the shipped lot (acceptance or rejection) should follow the sampling plan of MIL-STD-105E, single sampling, normal inspection, level II.

7-2 If the number of defects is equal to or less than the applicable acceptance level, the lot shall be accepted.

7-3 If the number of defects is more than the applicable acceptance level, the lot shall be rejected and the buyer should inform the seller of the result of incoming inspection in writing

8. Precaution:

Please pay attention to the following items when you use the LCD Module with back-light unit.

1. Do not twist or bend the module and prevent the unsuitable external force for display module during assembly.
2. Adopt measures for good heat radiation. Be sure to use the module with in the specified temperature.
3. Avoid dust or oil mist during assembly.
4. Follow the correct power sequence while operating. Do not apply the invalid signal, otherwise, it will cause improper shut down and damage the module.
5. Less EMI: it will be more safety and less noise.
6. Please operate module in suitable temperature. The response time & brightness will drift by different temperature.
7. Avoid displaying a fixed pattern (exclude the white pattern) for a long period, which may lead to image-sticking.
8. Be sure to turn off the power when connecting or disconnecting the circuit.
9. Polarizer scratches easily, please handle it carefully.
10. Display surface never likes dirt or stains.
11. A dewdrop may lead to destruction. Please wipe off any moisture before using module.
12. Sudden temperature changes cause condensation, and it will cause polarizer damaged.
13. High temperature and humidity may degrade performance. Please do not expose the module to the direct sunlight and so on.
14. Acetic acid or chlorine compounds are not friends with TFT display module.
15. Static electricity will damage the modules; please do not touch the module without any grounded device.
16. No parts of the panel should be dismantled without the presence of AU engineer or the product will not be warranted.
17. Be careful do not touch the rear side directly because of the backlight high voltage.
18. No strong vibration or shock. It will cause module broken.
19. Storage the modules in suitable environment with regular packing.
20. Be careful of injury from a broken display module. Please avoid the pressure adding to the surface (front or rear side) of modules, because it will cause the display non-uniformity or other function issue.
21. AHVA (Advanced-Hyper-Viewing-Angle) is a technology for LCD displays that offers users wider viewing angles and bolder colors by allowing more light to pass through the panel. As a result, AHVA panels can exhibit a slight glow around the edges and corners (also known as light bleed), which is entirely normal.
22. If customer found the visual defect panel during IQC inspection, please follow RMA rule that customer must provide defect panel SN & Carton number. If not, AUO can’t apply RMA process.

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